# Core practical – Investigating neutralisation

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

**Name** _____ **Class** _____ **Date** _____

## Practical Objective

Powdered calcium hydroxide reacts with hydrochloric acid. Calcium chloride solution and water are produced:

$$Ca(OH)_2\,(s) + 2HCl\,(aq) \rightarrow CaCl_2\,(aq) + 2H_2O\,(l)$$

You will investigate what happens to the pH of a fixed volume of dilute hydrochloric acid when you add calcium hydroxide to it.

## Apparatus

- eye protection
- 100 $cm^3$ beaker
- 50 $cm^3$ measuring cylinder
- ±0.1 g balance
- piece of paper or a 'weighing boat'
- spatula
- stirring rod
- white tile
- universal indicator paper
- pH colour chart
- dilute hydrochloric acid
- calcium hydroxide powder
- graph paper

## Safety

- Wear eye protection. Calcium hydroxide is an irritant with a risk of serious damage to eyes. Dilute hydrochloric acid is an irritant.

**Your teacher may watch to see if you can:**

- carry out an experiment appropriately.
- use apparatus accurately and safely.

## Method

**A** Use the measuring cylinder to add 50 $cm^3$ of dilute hydrochloric acid to the beaker.

**B** Estimate and record the pH of the contents of the beaker, as follows:

- Put a piece of universal indicator paper onto the white tile.
- Dip the end of the glass rod into the liquid, then tap it onto the universal indicator paper.
- Wait 30 seconds, then match the colour to the appropriate pH on the pH colour chart.
- Rinse the glass rod with water.

**C** Measure out 0.3 g of calcium hydroxide powder onto a piece of paper or a 'weighing boat'.

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**D** Add the calcium hydroxide powder to the beaker, stir, then estimate and record the pH of the mixture (as in step **B**).

**E** Repeat step **D** seven times so that you add a total of 2.4 g of calcium hydroxide powder to the acid.

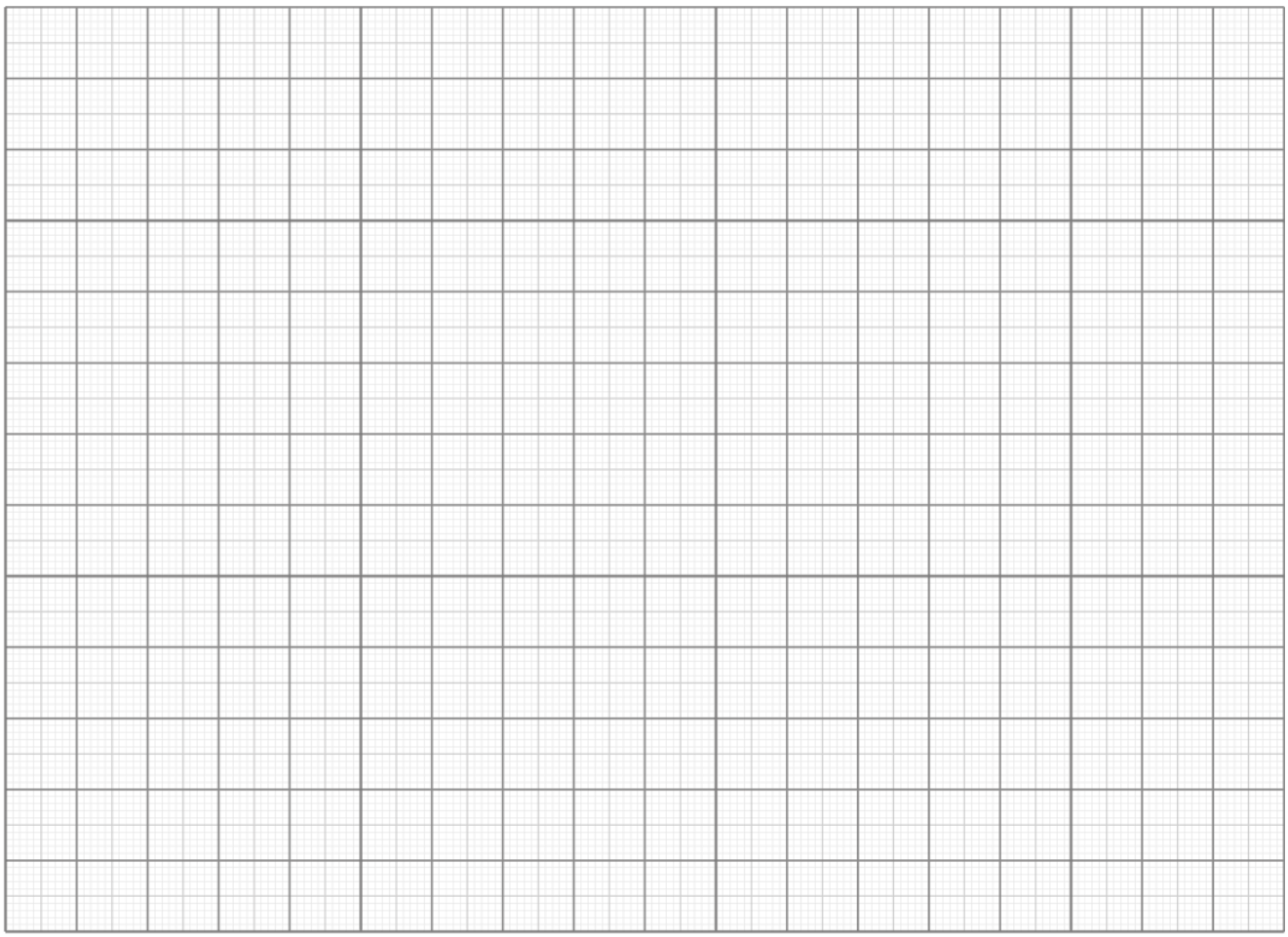
## Recording your results

**1** In the space below, make a table to record the pH of the contents of the beaker. Use columns for the mass of calcium hydroxide powder added, and the pH of the mixture. Remember to leave a row for the first pH measurement (before you have added any calcium hydroxide).

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## Considering your results

**2** Plot a line graph to show pH on the vertical axis and mass of calcium hydroxide added on the horizontal axis. Draw a curve of best fit.



**3** Describe what happens to the pH of the reaction mixture as calcium hydroxide continues to be added.

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**4** Use your graph to determine the mass of calcium hydroxide that must be added to reach pH 7.

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## Evaluation

**5** Explain one way to improve the accuracy of the experiment.

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